"What Is?" Seminar

UCP maps

Ran Kiri

Technion - Israel Istitute of Technology

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Let V be a subspace of $M_n(\mathbb{C})$, the space of complex $n \times n$ matrices, and let U be a subspace of $M_k(\mathbb{C})$.
There are many different ways to answer the question - when are V and U "the same"? For example:

- **As vector spaces** - if there is a linear bijection $T : V \to U$.
- **As normed space** - if there is a linear bijection $T : V \to U$ such that for all $x \in \mathbb{C}^n$:

$$\|Tx\| = \|x\|$$

- **As inner product spaces** - if there exists a linear bijection $T : V \to U$ such that for all $x, y \in V$:

$$\langle Tx, Ty \rangle = \langle x, y \rangle$$

Each bijection is chosen in such a way that certein quantities will be invariant under these maps (e.g. dimension, lengths, eigenvalues).

In this talk

We present another important structure and its natural maps - which is the partial ordering on PSD matrices, and unital and completely positive maps.

- If $\mathcal{H}$ is a complete inner product space (e.g. $\mathbb{C}^n$ with it's standard inner product), we denote by $B(\mathcal{H})$ as the collection of all linear operators $T : \mathcal{H} \to \mathcal{H}$ such that:

$$\|T\| := \sup_{\|h\|=1} \|Th\| < +\infty$$

- For any $T \in B(\mathcal{H})$, there exists a **unique** operator $T^* \in B(\mathcal{H})$, called the **adjoint operator**, such that for all $x, y \in \mathcal{H}$:

$$\langle Tx, y \rangle = \langle x, T^* y \rangle$$

- If $\mathcal{H} = \mathbb{C}^n$ with the standard inner product, $B(\mathcal{H})$ is just $M_n(\mathbb{C})$, and for any $A = (a_{ij})$, we have that $A^* = (\bar{a}_{ji})$.

The space $M_n(\mathbb{C})$

- $A \in M_n(\mathbb{C})$ is called **self-adjoint** if $A^* = A$,e.g. the matrix $\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$.
- $A \in M_n(\mathbb{C})$ is called **positive-semi-definite** if for all $x \in \mathbb{C}^n$, $\langle Ax, x \rangle \geq 0$.
  - In fact, A is PSD iff there exists some $B \in M_n(\mathbb{C})$ such that $A = B^*B$.For example:

$$\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^* \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$$

- For any $A \in M_n(\mathbb{C})$, we can write:

$$A = \frac{A + A^*}{2} + i\frac{A - A^*}{2i}$$

  which is a linear combination of two self adjoints.
- Without proof: any matrix can be written as the linear combination of 4 PSD matrices!

Since the entire matrix space is spanned by the positive matrices, they serve as a great tool to study the structure of $M_n(\mathbb{C})$.

But what structure can be studied with the self adjoints?

- We define a partial order on $M_n(\mathbb{C})$.We say that $A \geq B$ in $M_n(\mathbb{C})$ if $A - B$ is PSD.
- For example, $A = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix} \geq \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = B$ because:

$$A - B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \geq 0$$

- This partial order is obviously induced on any subspace.
- One special kind of subspaces are **Operator Systems**, which are self adjoint subspaces which contain the unit. (We will talk about them at the end of the talk).

So what are those order preserving maps?

Positive Maps

Assume $V \subset M_n$ is a subspace. A linear map $\phi : V \to \phi(V) \subset M_k$ is called **positive** if:

$$V \ni A \geq 0 \Longrightarrow \phi(V) \geq 0$$

Simple Example

Let $\phi : M_n(\mathbb{C}) \to M_1(\mathbb{C}) \cong \mathbb{C}$ be defined by $\phi(A) = \mathrm{tr}(A)$. Notice that A is PSD if and only if A is diagonalizeable with non-negative eigenvalues. Therefore if A is PSD, it has a non-negative trace.

Additional Structure and CP

We can introduce **even more structure**!
Let $V \subset M_n$ be a subspace. For any $k \in \mathbb{N}$, consider the space $M_k(V)$, which is the space of $k \times k$ matrices with elements from V (which are $n \times n$ matrices).By "erasing the braces", this is actually a subspace of M_{kn}.

CP and UCP maps

Let $V \in M_n$ be a subspace and let $\phi : V \to M_\ell$ a positive map.For any $k \in \mathbb{N}$, define $\phi_k : M_k(V) \to M_k(M_\ell)$, by:

$$\phi_k\left(\begin{pmatrix} v_{11} & \dots & v_{1k} \\ \vdots & \ddots & \vdots \\ v_{k1} & \dots & v_{kk} \end{pmatrix}\right) = \begin{pmatrix} \phi(v_{11}) & \dots & \phi(v_{1k}) \\ \vdots & \ddots & \vdots \\ \phi(v_{k1}) & \dots & \phi(v_{kk}) \end{pmatrix}$$

If ϕ_k is positive for all $k \in \mathbb{N}$, we say that ϕ is **Completely Positive** (CP).We also say that it is **unital** (UCP) if $\phi(I_n) = I_\ell$.

Additional Structure and CP

So as we can see, completely positive maps are "better".

- They preserve the matrix ordering.
- They preserve the ordering on "higher levels". Namely, the matricial order on $M_k(V)$ for all $k \in \mathbb{N}$.

But this might raise some questions:

- How do you check whether a map is completely positive?
- Are there even any completely positive maps?
- Why would a positive map not be completely positive?

As we will see in this talk, these questions have some pretty good and surprising answers.

UCP's and Conjugation

Assume B is any $k \times n$ matrix, and consider:

$$\phi : M_n \to M_k, \quad \phi(A) = B^* A B$$

- If $A = C^* C$ is PSD, then:

  $$\phi(A) = B^* C^* C B = (CB)^* (CB)$$

  which is also PSD. Therefore, ϕ is positive.
- Assume A is an element in $M_m(M_n)$ which is positive. Recall that this is an $m \times m$ matrix with entries from M_n, so it is of the form:

  $$A = \begin{pmatrix} A_{11} & \dots & A_{1m} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mm} \end{pmatrix}$$

  Such that $A_{ij} \in M_n$ for all i, j.

UCP's and Conjugation

- By the definition of ϕ_m:

$$\phi_m \begin{pmatrix} A_{11} & \dots & A_{1m} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mm} \end{pmatrix} = \begin{pmatrix} B^* A_{11} B & \dots & B^* A_{1m} B \\ \vdots & \ddots & \vdots \\ B^* A_{m1} B & \dots & B^* A_{mm} B \end{pmatrix}$$

  By considering block matrix multiplication, we can write:

$$= \begin{pmatrix} B & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & B \end{pmatrix}^* \begin{pmatrix} A_{11} & \dots & A_{1m} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mm} \end{pmatrix} \begin{pmatrix} B & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & B \end{pmatrix}$$

- Treating A as an $mn \times mn$ matrix, this is just a conjugation with a matrix, and therefore positive.

Corollary

Conjugations are completely positive maps! Moreover, conjugation by a **unitary** is UCP since $\phi(I_n) = U^* I_n U = I_n$.

Note - Since unitary conjugations preserve many structural aspects of matrix spaces, it is good to know that they are also UCP.

UCP's and *-Representations

A $*$-representation $\pi : V \to M_k$ is any linear map which is multiplicative and $*$-preserving.Meaning that:

$$\pi(AB) = \pi(A)\pi(B), \quad \pi(A^*) = \pi(A)^*$$

for all $A, B \in V$.

$*$-representations are CP

- π is positive - $\pi(A^*A) = \pi(A)^*\pi(A)$.
- $\pi_m : M_m(V) \to M_m(M_k)$ is also a $*$-representation:

$$\pi_m \begin{pmatrix} v_{11}^* & \dots & v_{m1}^* \\ \vdots & \ddots & \vdots \\ v_{1m}^* & \dots & v_{mm} \end{pmatrix} = \begin{pmatrix} \pi(v_{11}^*) & \dots & \pi(v_{m1}^*) \\ \vdots & \dots & \vdots \\ \pi(v_{1m}^*) & \dots & \pi(v_{mm})^* \end{pmatrix}$$

$$= \begin{pmatrix} \pi(v_{11})^* & \dots & \pi(v_{m1})^* \\ \vdots & \ddots & \vdots \\ \pi(v_{1m})^* & \dots & \pi(v_{mm})^* \end{pmatrix} = \pi_m \left(\begin{pmatrix} v_{11} & \dots & v_{1m} \\ \vdots & \ddots & \vdots \\ v_{m1} & \dots & v_{mm} \end{pmatrix} \right)^*$$

Non-CP maps

So far, any "standard" positive map we've encountered turned out to be completely positive. But unfortunately, this isn't always the case.

The Transpose Map

Define $\phi : M_2 \to M_2$ by $\phi(A) = A^t$.

- ϕ **is positive** - if $A = B^*B \in M_n$ is PSD, then:

$$\phi(A) = \phi(B^*B) = (B^*B)^t = B^t(B^*)^t = B^t\left(B^t\right)^* \geq 0$$

- ϕ **is not 2-positive** - since:

$$\phi_2\left(\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}\right) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

  This matrix has -1 as an eigenvalue, therefore - it is not PSD.

- In conclusion, ϕ is positive but not completely positive.

The Schur Product

Imagine, that in the first course in linear algebra, the lecturer would define matrix multiplication entry-wise:

$$\begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{n1} & \dots & b_{nn} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} & \dots & a_{1n}b_{1n} \\ \vdots & \ddots & \vdots \\ a_{1n}b_{1n} & \dots & a_{nn}b_{nn} \end{pmatrix}$$

How nice could it be?
This product, which is not our usual matrix multiplication, is called **the Schur Product** of A and B, and we denote it by $A * B$. As the next theorem shows, this map is a "good map".

Schur Product of A PSD Matrix

Let $S_A : M_n \to M_n$ be defined by $S_A(B) = A * B$. Then the following are equivalent:

1. A is PSD.
2. S_A is positive.
3. S_A is completely positive.

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3. S_A is completely positive.

Proof

- $(3) \implies (2)$ - Follows from the definition of CP.
- $(2) \implies (1)$ - Assume S_A is a positive map and consider $J = \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix}$.This matrix is seen easily to be positive, which means that $S_A(J) = A$ is PSD, which is what we wanted to prove.

The Schur Product

In order to give a simple proof for (1) $\Longrightarrow$ (3), we will "define" (loosely) the **Tensor product** of two vector spaces.

The Space $\mathbb{C}^n \otimes \mathbb{C}^m$

For a given $n, m \in \mathbb{N}$, denote by $\{e_i\}_{i=1}^n, \{e_i\}_{j=1}^m$ the standard basis vectors of $\mathbb{C}^n, \mathbb{C}^m$.

- The space $\mathbb{C}^n \otimes \mathbb{C}^m$ is the vector space:

$$\mathbb{C}^n \otimes \mathbb{C}^m := \left\{ \sum_{\substack{i=1,\ldots,n \\ j=1,\ldots,m}} a_{ij} e_i \otimes e_j \,\middle|\, a_{ij} \in \mathbb{C} \right\} = \mathrm{Span}_{\mathbb{C}} \{e_i \otimes e_j\}_{\substack{i=1,\ldots,n \\ j=1,\ldots,m}}$$

- The inner product on this space is given by:

$$\langle e_i \otimes e_j, e_m \otimes e_\ell \rangle = \delta_{im}\delta_{k\ell}$$

- For any two operators $A \in M_n, B \in M_m$ one defines $A \otimes B$ as the operator which acts by:

$$(A \otimes B)(e_i \otimes e_j) = (Ae_i) \otimes (Be_j)$$

An Alternative Approach to Schur Products

Define $V : \mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^n$ by $Ve_i = e_i \otimes e_i$ (and extend linearly).
The operator $V^*(A \otimes B)V$ acts on $\mathbb{C}^n$, by:

$$\langle V^*(A \otimes B)Ve_i, e_j \rangle \overset{\substack{\text{Adjoint} \\ \text{Property}}}{=} \langle (A \otimes B)(e_i \otimes e_i), e_j \otimes e_j \rangle$$

$$\overset{\substack{\text{Product} \\ \text{Operators}}}{=} \langle Ae_i \otimes Be_i, e_j \otimes e_j \rangle = \langle Ae_i, e_j \rangle \langle Be_i, e_j \rangle = a_{ij} b_{ij}$$

This means that for all $x, y \in \mathbb{C}^n$, we get:

$$\langle V^*(A \otimes B)Vx, y \rangle = \langle (A * B)x, y \rangle$$

And therefore $V^*(A \otimes B)V = A * B$. We are now ready to prove the last part of the theorem.

Proof

- (1) $\Longrightarrow$ (3) - Assume A is PSD, which means we can write it as S^*S. For a given $k \in \mathbb{N}$, consider the map $\pi : M_n \to M_n(M_n)$ given by:

$$\pi(B) = I_n \otimes B$$

Which is a $*$-representation. Now, note that we can write:

$$(S_A)_k \left(\begin{pmatrix} B_{11} & \dots & B_{1k} \\ \vdots & \ddots & \vdots \\ B_{k1} & \dots & B_{kk} \end{pmatrix} \right) = \begin{pmatrix} A * B_{11} & \dots & A * B_{1k} \\ \vdots & \ddots & \vdots \\ A * B_{k1} & \dots & A * B_{kk} \end{pmatrix}$$

$$= \begin{pmatrix} V^*(A \otimes B_{11})V & \dots & V^*(A \otimes B_{1k})V \\ \vdots & \ddots & \vdots \\ V^*(A \otimes B_{k1})V & \dots & V^*(A \otimes B_{kk})V \end{pmatrix}$$

Almost done!

Proof

$$= \begin{pmatrix} V^*(S^*S \otimes B_{11})V & \dots & V^*(S^*S \otimes B_{1k})V \\ \vdots & \ddots & \vdots \\ V^*(S^*S \otimes B_{k1})V & \dots & V^*(S^*S \otimes B_{kk})V \end{pmatrix}$$

Now, note that for all i, j, we get:

$$S^*S \otimes B_{ij} = (S \otimes I_n)^*(I_n \otimes B_{ij})(S \otimes I_n) = \tilde{S}^*\pi(B_{ij})\tilde{S}$$

Therefore this map can be written as:

$$= \begin{pmatrix} \tilde{S}V & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \tilde{S}V \end{pmatrix}^* \pi_k(B) \begin{pmatrix} \tilde{S}V & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \tilde{S}V \end{pmatrix}$$

which is a composition of a $*$-representation with a conjugation map, and therefore positive!

Using the Schur Product

Note that the theorem give us a nice way to check whether a matrix is PSD.For Example, note that for any $\lambda \geq 1$, the matrix:

$$D_\lambda = \begin{pmatrix} \lambda & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & \lambda \end{pmatrix} = J + (\lambda - 1)I_n$$

is positive. This is true because $D_\lambda \geq J \geq 0$. Therefore, for any A:

$$A \geq 0 \Longrightarrow D_\lambda * A \geq 0, \quad D_\lambda * A \geq 0$$

This means that multiplying the diagonal elements by some $\lambda \geq 1$ preserves positivity.[1] For an explicit example:

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \geq 0 \Longrightarrow \begin{pmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{pmatrix} \geq 0$$

[1] The even bigger surprise, is that because this map is CP, we can also multiply the diagonal of **each block** in the same way and preserve positivity.

Using the Schur Product

Another nice example is a necessary condition for a matrix to be PSD.By taking the Schur product with the identity I_n, we have that whenever $A \geq 0$:

$$I_n * A = \begin{pmatrix} a_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & a_n \end{pmatrix} \geq 0$$

Therefore, if a diagonal element of A is negative, A cannot be PSD.
For one final example, for $m < n$, denote by J_m the matrix whose ij-th entries are 1 for $i, j \leq m$ and 0 otherwise.If A is PSD, $J_m * A$ must also be PSD, but:

$$J_m * A = \begin{pmatrix} a_{11} & \dots & a_{1m} & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mm} & 0 & \dots & 0 \\ 0 & \ddots & 0 & 0 & \ddots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}$$

This shows that if one of this blocks is not positive, A cannot be PSD!

The Schur Product with a PSD matrix is completely positive, since it turned out to be a composition of a conjugation map with a $*$-representation.The next remarkable theorem by Stinespring, proves this is not a coincidence.

Stinespring Representation Theorem

Let $\mathcal{A}$ be a unital C^*-algebra (M_n for example) and let $\phi : \mathcal{A} \to B(\mathcal{H})$ be a completely positive map.
Then, there exists a Hilbert space $\mathcal{K}$, a unital $*$-representation $\pi : \mathcal{A} \to B(\mathcal{K})$ and a bounded operator $V : \mathcal{H} \to \mathcal{K}$, such that:

$$\phi(a) = V^* \pi(a) V$$

for all $a \in \mathcal{A}$. Moreover, $\|V\|^2 = \|\phi(1)\|$.

Therefore, **any positive map** from such C^*-algebras (such as M_n) to an operator space, is a composition of a conjugation and a representation. This completely characterizes these maps!

What Do You Do With CP-Maps?

Since we've seen that CP (and more specifically UCP) maps are the maps whose main property is order preserving, we want to use such maps to describe when are two operator systems "same" with respect to matricial ordering.

Complete Order Equivalence

Let $\mathcal{S}_1, \mathcal{S}_2$ be two operator systems. We say that they are **completely order equivalent**, if there exists a bijection $\phi : \mathcal{S}_1 \to \mathcal{S}_2$ such that ϕ is UCP and it's inverse is also UCP.

More specifically for any two d-tuple of operators $(T_1, \ldots, T_d), (T_1', \ldots, T_d')$ we say that these tuples are completely order equivalent, if:

$$\mathrm{Span}\{1, T_1, T_1^*, \ldots, T_d, T_d^*\} \cong_{\mathrm{ord}} \mathrm{Span}\{1, T_1', (T_1')^*, \ldots, T_d', (T_d')^*\}$$

And that map, ϕ, satisfies $\phi(T_i) = T_i'$.

We say that $(U_1, \ldots, U_m)$ is an m-tuple of spin unitaries, if each U_i is unitary and for all i, j:

$$U_i U_j = -U_j U_i$$

Example (Pauli Matrices)

For 2×2 matrices, the tuple:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is a spin tuple knows as the **Pauli Matrices**, which commonly arise in quantum mechanics.

Theorem (Farenick, Huntinghawk, Masanika, Plosker, 2020)

If $U = (U_1, \ldots, U_m), V = (V_1, \ldots, V_m)$ are m-tuples of spin unitaries, then $U \cong_{\text{ord}} V$.

COE of q-Commuting Unitaries

This interesting (and realitively new) theorem shows that the structure of spin tuples is uniquely determined by the cardinality of the tuple!
As an extension of that notion, we say that $T_1, \ldots, T_m$ are q commuting unitaries provided that they satisfy $T_i T_{i+1} = q T_{i+1} T_i$ and $q = e^{i\theta}$.

Theorem (Farenick, Ojo, Plosker, 2020)

Assume U, V and U', V' are two pairs of q-commuting unitaries, which are of p-th order, namely:

$$U^p = V^p = (U')^p = (V')^p = 1$$

Then, $(U, V) \cong_{\text{ord}} (U', V')$.

Unfortunately for $g \geq 3$, the same article found examples of g-tuples which are not completely order equivalent.
These q-commuting unitaries rise naturally in quantum mechanics and in the study of operator dilations, so understanding their mathematical structure is still an active area of research.

Questions?